

NTA JEE Mains Jan 2026

Test Date	24/01/2026
Test Time	3:00 PM - 6:00 PM
Subject	B. Tech

Section : Mathematics Section A

Q.1 Let $y = y(x)$ be a differentiable function in the interval $(0, \infty)$ such that $y(1) = 2$,

$$\text{and } \lim_{t \rightarrow x} \left(\frac{t^2 y(x) - x^2 y(t)}{x - t} \right) = 3 \text{ for each } x > 0. \text{ Then } 2y(2) \text{ is equal to}$$

- Options 1. 18
2. 27
3. 23
4. 12

Question Type : MCQ

Question ID : 444792620

Option 1 ID : 4447922119

Option 2 ID : 4447922120

Option 3 ID : 4447922117

Option 4 ID : 4447922118

Q.2 Let $\vec{a} = 2\hat{i} - 5\hat{j} + 5\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + 3\hat{k}$. If $\vec{c}$ is a vector such that

$$2(\vec{a} \times \vec{c}) + 3(\vec{b} \times \vec{c}) = \vec{0} \text{ and } (\vec{a} - \vec{b}) \cdot \vec{c} = -97, \text{ then } |\vec{c} \times \hat{k}|^2 \text{ is equal to}$$

- Options 1. 193
2. 218
3. 233
4. 205

Question Type : MCQ

Question ID : 444792615

Option 1 ID : 4447922097

Option 2 ID : 4447922099

Option 3 ID : 4447922100

Option 4 ID : 4447922098

Q.3 The letters of the word "UDAYPUR" are written in all possible ways with or without meaning and these words are arranged as in a dictionary. The rank of the word "UDAYPUR" is

- Options 1. 1580
2. 1581
3. 1579
4. 1578

Question Type : MCQ

Question ID : 444792608

Option 1 ID : 4447922071

Option 2 ID : 4447922072

Option 3 ID : 4447922070

Option 4 ID : 4447922069

Q.4 Let $[t]$ denote the greatest integer less than or equal to t . If the function

$$f(x) = \begin{cases} b^2 \sin \left(\frac{\pi}{2} \left[\frac{\pi}{2} (\cos x + \sin x) \cos x \right] \right), & x < 0 \\ \frac{\sin x - \frac{1}{2} \sin 2x}{x^3}, & x > 0 \\ a, & x = 0 \end{cases}$$

is continuous at $x = 0$, then $a^2 + b^2$ is equal to

Options

1. $\frac{5}{8}$
2. $\frac{3}{4}$
3. $\frac{9}{16}$
4. $\frac{1}{2}$

Question Type : MCQ

Question ID : 444792616

Option 1 ID : 4447922104

Option 2 ID : 4447922101

Option 3 ID : 4447922102

Option 4 ID : 4447922103

Q.5 Let a_1, a_2, a_3, a_4 be an A.P. of four terms such that each term of the A.P. and its common difference l are integers. If $a_1 + a_2 + a_3 + a_4 = 48$ and

$a_1 a_2 a_3 a_4 + l^4 = 361$, then the largest term of the A.P. is equal to

Options

1. 23
2. 24
3. 27
4. 21

Question Type : MCQ

Question ID : 444792605

Option 1 ID : 4447922058

Option 2 ID : 4447922059

Option 3 ID : 4447922060

Option 4 ID : 4447922057

Q.6 Consider the following three statements for the function $f: (0, \infty) \rightarrow \mathbb{R}$ defined by

$$f(x) = |\log_e x| - |x - 1|;$$

(I) f is differentiable at all $x > 0$.

(II) f is increasing in $(0, 1)$.

(III) f is decreasing in $(1, \infty)$.

Then.

- Options
1. Only (I) and (III) are TRUE.
 2. Only (II) and (III) are TRUE.
 3. All (I), (II) and (III) are TRUE.
 4. Only (I) is TRUE.

Question Type : MCQ

Question ID : 444792617

Option 1 ID : 4447922107

Option 2 ID : 4447922106

Option 3 ID : 4447922108

Option 4 ID : 4447922105

Q.7 Let the angles made with the positive x -axis by two straight lines drawn from the point $P(2, 3)$ and meeting the line $x + y = 6$ at a distance $\sqrt{\frac{2}{3}}$ from the point P be

θ_1 and θ_2 . Then the value of $(\theta_1 + \theta_2)$ is:

- Options
1. $\frac{\pi}{12}$
 2. $\frac{\pi}{3}$
 3. $\frac{\pi}{6}$
 4. $\frac{\pi}{2}$

Question Type : MCQ

Question ID : 444792612

Option 1 ID : 4447922088

Option 2 ID : 4447922085

Option 3 ID : 4447922087

Option 4 ID : 4447922086

Q.8 The largest value of n , for which 40^n divides $60!$, is

- Options
1. 14
 2. 11
 3. 12
 4. 13

Question Type : MCQ

Question ID : 444792607

Option 1 ID : 4447922067

Option 2 ID : 4447922068

Option 3 ID : 4447922065

Option 4 ID : 4447922066

Q.9 Let the image of parabola $x^2 = 4y$, in the line $x - y = 1$ be $(y + a)^2 = b(x - c)$, $a, b, c \in \mathbb{N}$. Then $a + b + c$ is equal to

- Options 1. 6
2. 4
3. 12
4. 8

Question Type : MCQ

Question ID : 444792611

Option 1 ID : 4447922082

Option 2 ID : 4447922081

Option 3 ID : 4447922084

Option 4 ID : 4447922083

Q.10 The smallest positive integral value of a , for which all the roots of $x^4 - ax^2 + 9 = 0$ are real and distinct, is equal to

- Options 1. 3
2. 9
3. 7
4. 4

Question Type : MCQ

Question ID : 444792603

Option 1 ID : 4447922049

Option 2 ID : 4447922052

Option 3 ID : 4447922050

Option 4 ID : 4447922051

Q.11 Let $X = \{x \in \mathbb{N} : 1 \leq x \leq 19\}$ and for some $a, b \in \mathbb{R}$, $Y = \{ax + b : x \in X\}$. If the mean and variance of the elements of Y are 30 and 750, respectively, then the sum of all possible values of b is

- Options 1. 100
2. 60
3. 80
4. 20

Question Type : MCQ

Question ID : 444792609

Option 1 ID : 4447922076

Option 2 ID : 4447922074

Option 3 ID : 4447922075

Option 4 ID : 4447922073

Q.12 Let $\vec{a} = 2\hat{i} - \hat{j} - \hat{k}$, $\vec{b} = \hat{i} + 3\hat{j} - \hat{k}$ and $\vec{c} = 2\hat{i} + \hat{j} + 3\hat{k}$. Let $\vec{v}$ be the vector in the plane of the vectors $\vec{a}$ and $\vec{b}$, such that the length of its projection on the vector $\vec{c}$ is $\frac{1}{\sqrt{14}}$. Then $|\vec{v}|$ is equal to

Options 1. 13

2. 7

3. $\frac{\sqrt{35}}{2}$

4. $\frac{\sqrt{21}}{2}$

Question Type : MCQ

Question ID : 444792614

Option 1 ID : 4447922094

Option 2 ID : 4447922093

Option 3 ID : 4447922096

Option 4 ID : 4447922095

Q.13 Let the length of the latus rectum of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, ($a > b$), be 30. If its eccentricity is the maximum value of the function $f(t) = -\frac{3}{4} + 2t - t^2$, then $(a^2 + b^2)$ is equal to

Options 1. 256

2. 276

3. 516

4. 496

Question Type : MCQ

Question ID : 444792610

Option 1 ID : 4447922077

Option 2 ID : 4447922080

Option 3 ID : 4447922079

Option 4 ID : 4447922078

Q.14 $\left(\frac{1}{3} + \frac{4}{7}\right) + \left(\frac{1}{3^2} + \frac{1}{3} \times \frac{4}{7} + \frac{4^2}{7^2}\right) + \left(\frac{1}{3^3} + \frac{1}{3^2} \times \frac{4}{7} + \frac{1}{3} \times \frac{4^2}{7^2} + \frac{4^3}{7^3}\right) + \dots$ upto infinite terms,
is equal to

- Options
1. $\frac{6}{5}$
 2. $\frac{4}{3}$
 3. $\frac{5}{2}$
 4. $\frac{7}{4}$

Question Type : MCQ

Question ID : 444792606

Option 1 ID : 4447922061

Option 2 ID : 4447922062

Option 3 ID : 4447922063

Option 4 ID : 4447922064

Q.15 Let $P = [p_{ij}]$ and $Q = [q_{ij}]$ be two square matrices of order 3 such that
 $q_{ij} = 2^{(i+j-1)} p_{ij}$ and $\det(Q) = 2^{10}$. Then the value of $\det(\text{adj}(\text{adj } P))$ is:

- Options
1. 16
 2. 81
 3. 32
 4. 124

Question Type : MCQ

Question ID : 444792604

Option 1 ID : 4447922053

Option 2 ID : 4447922054

Option 3 ID : 4447922055

Option 4 ID : 4447922056

Q.16 The sum of all values of α , for which the shortest distance between the lines
 $\frac{x+1}{\alpha} = \frac{y-2}{-1} = \frac{z-4}{-\alpha}$ and $\frac{x}{\alpha} = \frac{y-1}{2} = \frac{z-1}{2\alpha}$ is $\sqrt{2}$, is

- Options
1. 8
 2. -8
 3. -6
 4. 6

Question Type : MCQ

Question ID : 444792613

Option 1 ID : 4447922092

Option 2 ID : 4447922091

Option 3 ID : 4447922089

Option 4 ID : 4447922090

Q.17

Let $f(x) = \int \frac{7x^{10} + 9x^8}{(1+x^2+2x^9)^2} dx, x > 0, \lim_{x \rightarrow 0} f(x) = 0$ and $f(1) = \frac{1}{4}$.

If $A = \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{4} & f'(1) & 1 \\ \alpha^2 & 4 & 1 \end{bmatrix}$ and $B = \text{adj}(\text{adj } A)$ be such that $|B| = 81$, then α^2 is

equal to

Options 1. 3

2. 4

3. 2

4. 1

Question Type : MCQ

Question ID : 444792618

Option 1 ID : 4447922111

Option 2 ID : 4447922112

Option 3 ID : 4447922110

Option 4 ID : 4447922109

Q.18 Let $f(\alpha)$ denote the area of the region in the first quadrant bounded by $x = 0, x = 1, y^2 = x$ and $y = |\alpha x - 5| - |1 - \alpha x| + \alpha x^2$. Then $(f(0) + f(1))$ is equal to

Options 1. 7

2. 12

3. 9

4. 14

Question Type : MCQ

Question ID : 444792619

Option 1 ID : 4447922113

Option 2 ID : 4447922115

Option 3 ID : 4447922114

Option 4 ID : 4447922116

Q.19

If the domain of the function $f(x) = \sin^{-1}\left(\frac{1}{x^2 - 2x - 2}\right)$ is $(-\infty, \alpha] \cup [\beta, \gamma] \cup [\delta, \infty)$, then $\alpha + \beta + \gamma + \delta$ is equal to

Options 1. 3

2. 5

3. 4

4. 2

Question Type : MCQ

Question ID : 444792601

Option 1 ID : 4447922042

Option 2 ID : 4447922044

Option 3 ID : 4447922043

Option 4 ID : 4447922041

Q.20

Let f be a function such that $3f(x) + 2f\left(\frac{m}{19x}\right) = 5x$, $x \neq 0$, where

$$m = \sum_{i=1}^9 (i)^2. \text{ Then } f(5) - f(2) \text{ is equal to}$$

Options 1. 36

2. 9

3. -9

4. 18

Question Type : MCQ

Question ID : 444792602

Option 1 ID : 4447922048

Option 2 ID : 4447922046

Option 3 ID : 4447922045

Option 4 ID : 4447922047

Section : Mathematics Section B

Q.21 Let (h, k) lie on the circle $C : x^2 + y^2 = 4$ and the point $(2h + 1, 3k + 2)$ lie on an ellipse with eccentricity e . Then the value of $\frac{5}{e^2}$ is equal to _____.

Question Type : SA

Question ID : 444792623

Q.22 If $f(x)$ satisfies the relation $f(x) = e^x + \int_0^1 (y + xe^x) f(y) dy$, then $e + f(0)$ is equal to _____.

Question Type : SA

Question ID : 444792625

Q.23 The number of elements in the set $\{x \in [0, 180^\circ] : \tan(x + 100^\circ) = \tan(x + 50^\circ) \tan x \tan(x - 50^\circ)\}$ is _____.

Question Type : SA

Question ID : 444792624

Q.24 Let S be a set of 5 elements and $P(S)$ denote the power set of S . Let E be an event of choosing an ordered pair (A, B) from the set $P(S) \times P(S)$ such that $A \cap B = \emptyset$. If the probability of the event E is $\frac{3^p}{2^q}$, where $p, q \in \mathbb{N}$, then $p + q$ is equal to _____

Question Type : SA

Question ID : 444792622

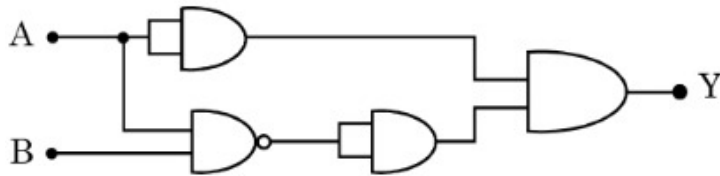
Q.25 Let $z = (1 + i)(1 + 2i)(1 + 3i) \dots (1 + ni)$, where $i = \sqrt{-1}$. If $|z|^2 = 44200$, then n is equal to _____

Question Type : SA

Question ID : 444792621

Section : Physics Section A

Q.26 Identify the correct truth table of the given logic circuit.



Options

1.

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

2.

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	0

3.

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

4.

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

Question Type : MCQ

Question ID : 444792644

Option 1 ID : 4447922198

Option 2 ID : 4447922200

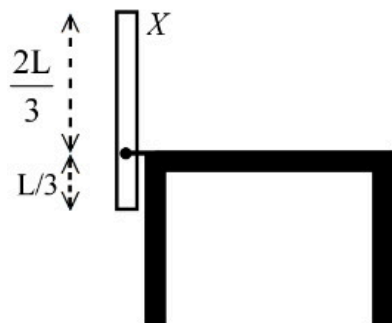
Option 3 ID : 4447922201

Option 4 ID : 4447922199

Q.27

A thin uniform rod (X) of mass M and length L is pivoted at a height $\left(\frac{L}{3}\right)$ as shown in the figure. The rod is allowed to fall from a vertical position and lie horizontally on the table. The angular velocity of this rod when it hits the table top, is _____.

(g = gravitational acceleration)



Options

1. $\sqrt{\frac{3g}{L}}$
2. $\sqrt{\frac{3}{2} \frac{g}{L}}$
3. $\frac{1}{\sqrt{2}} \sqrt{\frac{g}{L}}$
4. $\frac{3}{\sqrt{2}} \sqrt{\frac{g}{L}}$

Question Type : MCQ

Question ID : 444792627

Option 1 ID : 4447922132

Option 2 ID : 4447922130

Option 3 ID : 4447922133

Option 4 ID : 4447922131

Q.28 A point source is kept at the center of a spherically enclosed detector. If the volume of the detector increased by 8 times, the intensity will

- Options
1. decrease by 8 times
 2. decrease by 4 times
 3. increase by 8 times
 4. increase by 64 times

Question Type : MCQ

Question ID : 444792638

Option 1 ID : 4447922176

Option 2 ID : 4447922177

Option 3 ID : 4447922174

Option 4 ID : 4447922175

Q.29 The fifth harmonic of a closed organ pipe is found to be in unison with the first harmonic of an open pipe. The ratio of lengths of closed pipe to that of the open pipe is $5/x$. The value of x is _____.

- Options
- 1
 - 2
 - 4
 - 3

Question Type : **MCQ**

Question ID : **4447922633**

Option 1 ID : **4447922157**

Option 2 ID : **4447922154**

Option 3 ID : **4447922156**

Option 4 ID : **4447922155**

Q.30 Distance between an object and three times magnified real image is 40 cm. The focal length of the mirror used is _____ cm.

- Options
- 15
 - 20
 - 15/2
 - 10

Question Type : **MCQ**

Question ID : **4447922641**

Option 1 ID : **4447922188**

Option 2 ID : **4447922189**

Option 3 ID : **4447922187**

Option 4 ID : **4447922186**

Q.31 In a vernier callipers, 50 vernier scale divisions are equal to 48 main scale divisions. If one main scale division = 0.05 mm, then the least count of the vernier callipers is _____ mm.

- Options
- 0.02
 - 0.002
 - 0.05
 - 0.005

Question Type : **MCQ**

Question ID : **4447922626**

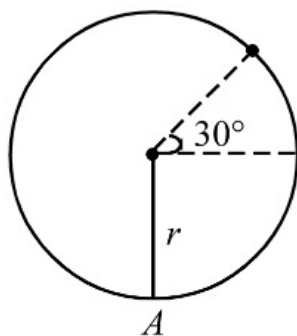
Option 1 ID : **4447922126**

Option 2 ID : **4447922127**

Option 3 ID : **4447922128**

Option 4 ID : **4447922129**

Q.32 In case of vertical circular motion of a particle by a thread of length r if the tension in the thread is zero at an angle 30° shown in figure, the velocity at the bottom point (A) of the circular path is
(g = gravitational acceleration)



- Options
1. $\sqrt{5gr}$
 2. $\sqrt{4gr}$
 3. $\sqrt{\frac{5}{2}gr}$
 4. $\sqrt{\frac{7}{2}gr}$

Question Type : **MCQ**

Question ID : **444792630**

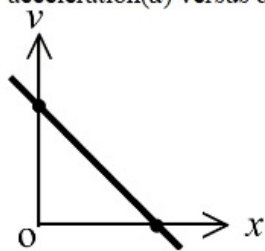
Option 1 ID : **4447922143**

Option 2 ID : **4447922145**

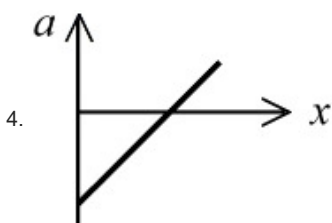
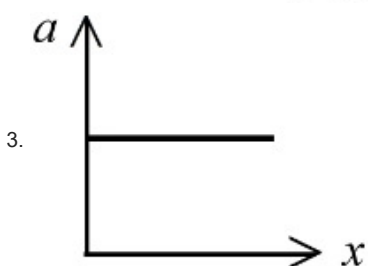
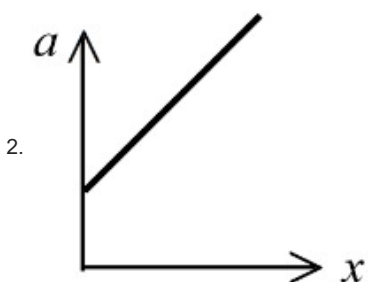
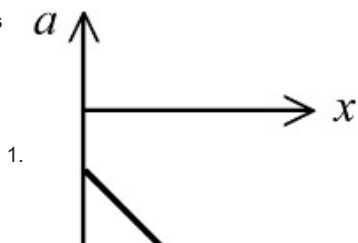
Option 3 ID : **4447922142**

Option 4 ID : **4447922144**

Q.33 The velocity (v) – Distance (x) graph is shown in figure. Which graph represents acceleration(a) versus distance (x) variation of this system?



Options



Question Type : MCQ

Question ID : 444792628

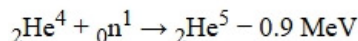
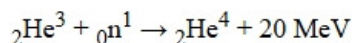
Option 1 ID : 4447922134

Option 2 ID : 4447922135

Option 3 ID : 4447922137

Option 4 ID : 4447922136

Q.34 The binding energy for the following nuclear reactions are expressed in MeV.



If X_3 , X_4 , X_5 denote the stability of ${}_2\text{He}^3$, ${}_2\text{He}^4$ and ${}_2\text{He}^5$, respectively, then the correct order is :

- Options
1. $X_4 = X_5 = X_3$
 2. $X_4 < X_5 < X_3$
 3. $X_4 > X_5 < X_3$
 4. $X_4 > X_5 > X_3$

Question Type : **MCQ**

Question ID : **444792643**

Option 1 ID : **4447922196**

Option 2 ID : **4447922194**

Option 3 ID : **4447922197**

Option 4 ID : **4447922195**

Q.35 When a light of a given wavelength falls on a metallic surface the stopping potential for photoelectrons is 3.2 V. If a second light having wavelength twice of first light is used, the stopping potential drops to 0.7 V. The wavelength of first light is _____ m.

($h = 6.63 \times 10^{-34} \text{ J.s}$, $e = 1.6 \times 10^{-19} \text{ C}$, $c = 3 \times 10^8 \text{ m/s}$)

- Options
1. 2.9×10^{-8}
 2. 2.5×10^{-7}
 3. 2.2×10^{-8}
 4. 3.1×10^{-7}

Question Type : **MCQ**

Question ID : **444792642**

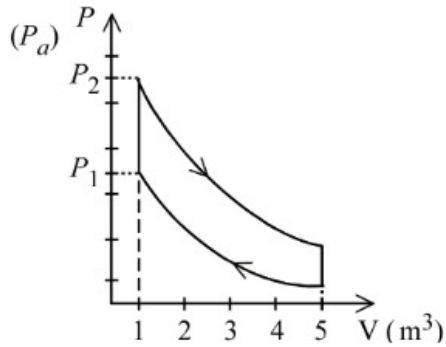
Option 1 ID : **4447922192**

Option 2 ID : **4447922191**

Option 3 ID : **4447922193**

Option 4 ID : **4447922190**

Q.36 10 mole of an ideal gas is undergoing the process shown in the figure. The heat involved in the process from P_1 to P_2 is α Joule ($P_1 = 21.7$ Pa and $P_2 = 30$ Pa, $C_v = 21$ J/K.mol, $R = 8.3$ J/mol.K). The value of α is _____.



- Options
1. 28
 2. 24
 3. 21
 4. 15

Question Type : MCQ

Question ID : 444792632

Option 1 ID : 4447922153

Option 2 ID : 4447922152

Option 3 ID : 4447922151

Option 4 ID : 4447922150

Q.37 Five persons P_1, P_2, P_3, P_4 and P_5 recorded object distance (u) and image distance (v) using same convex lens having power +5D as (25,96), (30,62), (35,37), (45,35) and (50,32) respectively. Identify correct statement

- Options
1. Readings recorded by all persons are correct
 2. Readings recorded by P_4 and P_5 persons are incorrect
 3. Readings recorded by P_3 and P_2 persons are incorrect
 4. Readings recorded by P_3 person are incorrect

Question Type : MCQ

Question ID : 444792639

Option 1 ID : 4447922178

Option 2 ID : 4447922181

Option 3 ID : 4447922180

Option 4 ID : 4447922179

Q.38 In the Young's double slit experiment the intensity produced by each one of the individual slits is I_0 . The distance between two slits is 2 mm . The distance of screen from slits is 10 m. The wavelength of light is $6000 \text{ \AA}$. The intensity of light on the screen in front of one of the slits is _____.

- Options
1. $2I_0$
 2. $4I_0$
 3. I_0
 4. $\frac{I_0}{2}$

Question Type : **MCQ**

Question ID : **444792640**

Option 1 ID : **4447922184**

Option 2 ID : **4447922185**

Option 3 ID : **4447922183**

Option 4 ID : **4447922182**

Q.39 A flexible chain of mass m hangs between two fixed points at the same level. The inclination of the chain with the horizontal at the two points of support is 30° . Considering the equilibrium of each half of the chain, the tension of the chain at the lowest point is _____.

- Options
1. $\frac{1}{2}mg$
 2. mg
 3. $\sqrt{3}mg$
 4. $\frac{\sqrt{3}}{2}mg$

Question Type : **MCQ**

Question ID : **444792629**

Option 1 ID : **4447922140**

Option 2 ID : **4447922141**

Option 3 ID : **4447922139**

Option 4 ID : **4447922138**

Q.40 A cubical block of density $\rho_b = 600 \text{ kg/m}^3$ floats in a liquid of density $\rho_e = 900 \text{ kg/m}^3$. If the height of block is $H = 8.0 \text{ cm}$ then height of the submerged part is _____ cm.

- Options
1. 5.3
 2. 7.3
 3. 4.3
 4. 6.3

Question Type : **MCQ**

Question ID : **444792631**

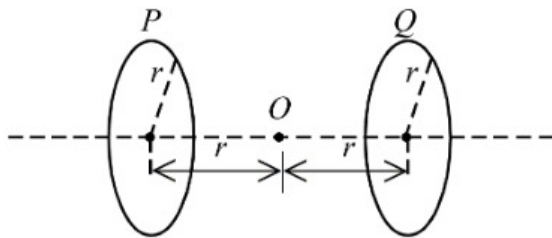
Option 1 ID : **4447922147**

Option 2 ID : **4447922149**

Option 3 ID : **4447922146**

Option 4 ID : **4447922148**

Q.41



Two identical circular loops P and Q each of radius r are lying in parallel planes such that they have common axis. The current through P and Q are I and $4I$ respectively in clockwise direction as seen from O . The net magnetic field at O is:

Options

1. $\frac{\mu_0 I}{4\sqrt{2}r}$ towards Q
2. $\frac{3\mu_0 I}{4\sqrt{2}r}$ towards P
3. $\frac{\mu_0 I}{4\sqrt{2}r}$ towards P
4. $\frac{3\mu_0 I}{4\sqrt{2}r}$ towards Q

Question Type : MCQ

Question ID : 444792635

Option 1 ID : 4447922165

Option 2 ID : 4447922162

Option 3 ID : 4447922164

Option 4 ID : 4447922163

Q.42 A regular hexagon is formed by six wires each of resistance $r \Omega$ and the corners are joined to the centre by wires of same resistance. If the current enters at one corner and leaves at the opposite corner, the equivalent resistance of the hexagon between the two opposite corners will be

Options

1. $\frac{3}{5}r$
2. $\frac{3}{4}r$
3. $\frac{5}{8}r$
4. $\frac{4}{5}r$

Question Type : MCQ

Question ID : 444792636

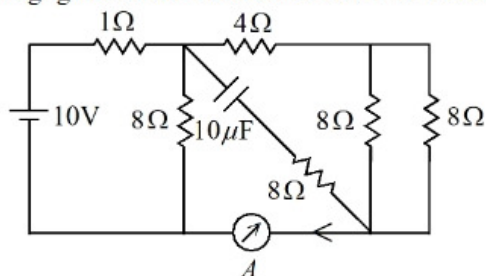
Option 1 ID : 4447922167

Option 2 ID : 4447922166

Option 3 ID : 4447922169

Option 4 ID : 4447922168

Q.43 The reading of the ammeter (A) in steady state in the following circuit (assuming negligible internal resistance of the ammeter) is _____ A.



Options 1. $\frac{1}{2}$

2. 2

3. 1

4. 0

Question Type : MCQ

Question ID : 444792645

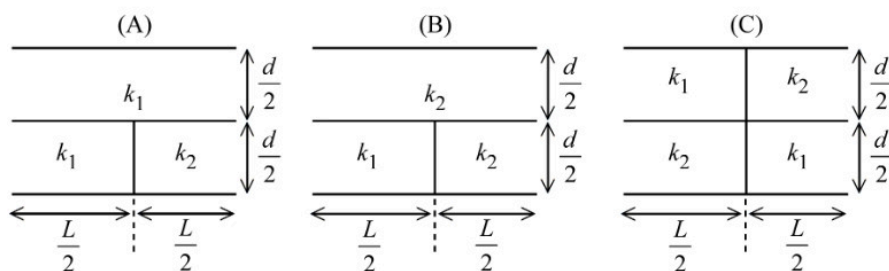
Option 1 ID : 4447922203

Option 2 ID : 4447922204

Option 3 ID : 4447922202

Option 4 ID : 4447922205

Q.44 Three parallel plate capacitors each with area A and separation d are filled with two dielectric (k_1 and k_2) in the following fashion. Which of the following is true? ($k_1 > k_2$)



Options 1. $C_C > C_B > C_A$

2. $C_A > C_C > C_B$

3. $C_C > C_A > C_B$

4. $C_B > C_C > C_A$

Question Type : MCQ

Question ID : 444792637

Option 1 ID : 4447922171

Option 2 ID : 4447922170

Option 3 ID : 4447922173

Option 4 ID : 4447922172

Q.45 A moving coil galvanometer of resistance $100\ \Omega$ shows a full scale deflection for a current of 1 mA . The value of resistance required to convert this galvanometer into an ammeter, showing full scale deflection for a current of 5 mA , is _____ Ω

- Options 1. 25
2. 10
3. 0.5
4. 2.5

Question Type : MCQ

Question ID : 444792634

Option 1 ID : 4447922161

Option 2 ID : 4447922160

Option 3 ID : 4447922159

Option 4 ID : 4447922158

Section : Physics Section B

Q.46 A soap bubble of surface tension 0.04 N/m is blown to a diameter of 7 cm . If $(15000 - x)\ \mu\text{J}$ of work is done in blowing it further to make its diameter 14 cm , then the value of x is _____.
($\pi = 22/7$)

Question Type : SA

Question ID : 444792648

Q.47 When 300 J of heat given to an ideal gas with $C_p = \frac{7}{2}R$ its temperature raises from 20°C to 50°C keeping its volume constant. The mass of the gas is (approximately) _____ g. ($R = 8.314\text{ J/mol.K}$)

Question Type : SA

Question ID : 444792647

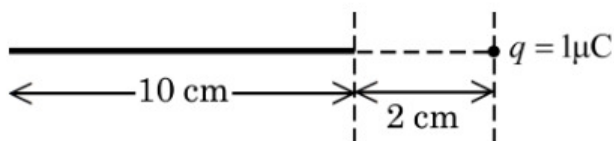
Q.48 A uniform solid cylinder of length L and radius R has moment of inertia about its axis equal to I_1 . A small co-centric cylinder of length $L/2$ and radius $R/3$ carved from this cylinder has moment of inertia about its axis equals to I_2 . The ratio I_1/I_2 is _____.

Question Type : SA

Question ID : 444792649

Q.49 A point charge $q = 1\ \mu\text{C}$ is located at a distance 2 cm from one end of a thin insulating wire of length 10 cm having a charge $Q = 24\ \mu\text{C}$, distributed uniformly along its length, as shown in figure. Force between q and wire is _____ N.

(Use : $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9\text{ N.m}^2/\text{C}^2$)



Question Type : SA

Question ID : 444792646

Q.50 In a meter bridge experiment to determine the value of unknown resistance, first the resistances $2\ \Omega$ and $3\ \Omega$ are connected in the left and right gaps of the bridge and the null point is obtained at a distance l cm from the left. Now when an unknown resistance $x\ \Omega$ is connected in parallel to $3\ \Omega$ resistance, the null point is shifted by 10 cm to the right of wire. The value of unknown resistance x is _____ Ω .

Question Type : SA

Question ID : 444792650

Section : Chemistry Section A

Q.51 Find out the statements which are **not** true.

- A. Resonating structures with more number of covalent bonds and lesser charge separation are more stable.
- B. In electromeric effect, an unsaturated system shows +E effect with nucleophile and -E effect with electrophile.
- C. Inductive effect is responsible for high melting point, boiling point and dipole moment of polar compounds.
- D. The greater the number of alkyl groups attached to the doubly bonded carbon atoms, higher is the heat of hydrogenation.
- E. Stability of carbanion increases with the increase in s - character of the carbon carrying the negative charge.

Choose the **correct** answer from the options given below:

- Options
- 1. A, C & D only
 - 2. B, D & E only
 - 3. A, D & E only
 - 4. B & D only

Question Type : MCQ

Question ID : 444792662

Option 1 ID : 4447922256

Option 2 ID : 4447922257

Option 3 ID : 4447922258

Option 4 ID : 4447922255

Q.52 A student has planned to prepare acetanilide from aniline using acetic anhydride.

The student has started from 9.3 g of aniline. However, the student has managed to obtain 11 g of dry acetanilide.

The % yield of this reaction is :-

- Options
- 1. 81.5%
 - 2. 59.5%
 - 3. 72.5%
 - 4. 97.5%

Question Type : MCQ

Question ID : 444792666

Option 1 ID : 4447922272

Option 2 ID : 4447922274

Option 3 ID : 4447922271

Option 4 ID : 4447922273

Q.53 The number of possible tripeptides formed involving alanine (ala), glycine (gly) and valine (val), where no amino acid has been used more than once is:

- Options
1. 8
 2. 3
 3. 6
 4. 4

Question Type : **MCQ**

Question ID : **4447922669**

Option 1 ID : **4447922286**

Option 2 ID : **4447922283**

Option 3 ID : **4447922285**

Option 4 ID : **4447922284**

Q.54 Two liquids A and B form an ideal solution at temperature T K. At T K, the vapour pressures of pure A and B are 55 and 15 kN m⁻² respectively. What is the mole fraction of A in solution of A and B in equilibrium with a vapour in which the mole fraction of A is 0.8?

- Options
1. 0.5217
 2. 0.663
 3. 0.480
 4. 0.340

Question Type : **MCQ**

Question ID : **4447922656**

Option 1 ID : **4447922233**

Option 2 ID : **4447922232**

Option 3 ID : **4447922231**

Option 4 ID : **4447922234**

Q.55 Pair of species among the following having same bond order as well as paramagnetic character will be-

- Options
1. O₂⁻, N₂⁺
 2. O₂⁺, N₂²⁻
 3. O₂⁺, N₂⁻
 4. O₂⁻, N₂⁻

Question Type : **MCQ**

Question ID : **4447922653**

Option 1 ID : **4447922219**

Option 2 ID : **4447922221**

Option 3 ID : **4447922222**

Option 4 ID : **4447922220**

Q.56 At 298 K, the mole percentage of $\text{N}_2(\text{g})$ in air is 80%. Water is in equilibrium with air at a pressure of 10 atm. What is the mole fraction of $\text{N}_2(\text{g})$ in water at 298 K?

(K_{H} for N_2 is 6.5×10^7 mm Hg)

- Options
1. 1.17×10^{-4}
 2. 1.23×10^{-7}
 3. 9.35×10^5
 4. 9.35×10^{-5}

Question Type : **MCQ**

Question ID : **444792655**

Option 1 ID : **4447922230**

Option 2 ID : **4447922228**

Option 3 ID : **4447922229**

Option 4 ID : **4447922227**

Q.57 Choose the **INCORRECT** statement

Options

1. CO_2 is the most acidic oxide among the dioxides of group of 14 elements.

2. Among the isotopes of carbon, ^{13}C is a radioactive isotope.

3. Carbon exhibits negative oxidation states along with +4 and +2.

4. Carbon cannot exceed its covalency more than four.

Question Type : **MCQ**

Question ID : **444792659**

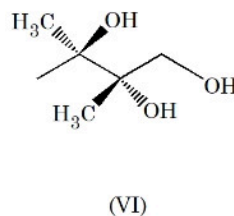
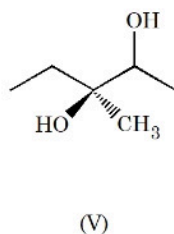
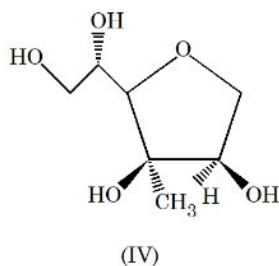
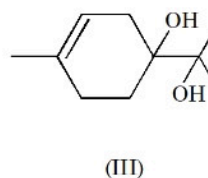
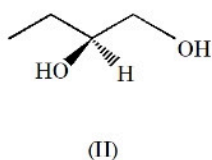
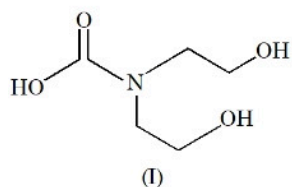
Option 1 ID : **4447922246**

Option 2 ID : **4447922245**

Option 3 ID : **4447922244**

Option 4 ID : **4447922243**

Q.58 From the following, how many compounds contain at least one secondary alcohol?



Choose the **correct** answer from the options given below:

- Options
1. Three
 2. Two
 3. Four
 4. Five

Question Type : **MCQ**

Question ID : **444792667**

Option 1 ID : **4447922275**

Option 2 ID : **4447922277**

Option 3 ID : **4447922278**

Option 4 ID : **4447922276**

Q.59 In the Group analysis of cations, Ba^{2+} & Ca^{2+} are precipitated respectively as

- Options
1. sulphide & sulphide
 2. carbonate & carbonate
 3. hydroxide & carbonate
 4. chromate & sulphide

Question Type : **MCQ**

Question ID : **444792670**

Option 1 ID : **4447922287**

Option 2 ID : **4447922289**

Option 3 ID : **4447922288**

Option 4 ID : **4447922290**

Q.60 The heat of atomisation of methane and ethane are 'x' kJ mol⁻¹ and 'y' kJ mol⁻¹ respectively. The longest wavelength (λ) of light capable of breaking the C-C bond can be expressed in SI unit as:

- Options
1. $\frac{N_A hc}{250(4y - 6x)}$
 2. $N_A hc \left(y - \frac{6x}{4} \right)^{-1}$
 3. $\frac{N_A hc}{250(y - 6x)}$
 4. $\frac{hc}{1000} \left(\frac{y - 6x}{4} \right)^{-1}$

Question Type : MCQ

Question ID : 444792654

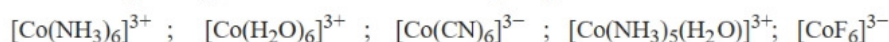
Option 1 ID : 4447922223

Option 2 ID : 4447922226

Option 3 ID : 4447922225

Option 4 ID : 4447922224

Q.61 The wavelength of light absorbed for the following complexes are in the order



(I)

(II)

(III)

(IV)

(V)

- Options
1. III < I < IV < II < V
 2. III < IV < I < II < V
 3. III < I < IV < V < II
 4. III < I < II < IV < V

Question Type : MCQ

Question ID : 444792661

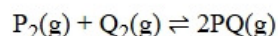
Option 1 ID : 4447922253

Option 2 ID : 4447922251

Option 3 ID : 4447922254

Option 4 ID : 4447922252

Q.62 Consider the following gaseous equilibrium in a closed container of volume 'V' at T(K).



2 moles each of $\text{P}_2(\text{g})$, $\text{Q}_2(\text{g})$ and $\text{PQ}(\text{g})$ are present at equilibrium. Now one mole each of ' P_2 ' and ' Q_2 ' are added to the equilibrium keeping the temperature at T(K).

The number of moles of P_2 , Q_2 and PQ at the new equilibrium, respectively, are

- Options
1. 2.67, 2.67, 2.67
 2. 2.56, 1.62, 2.24
 3. 1.21, 2.24, 1.56
 4. 1.66, 1.66, 1.66

Question Type : MCQ

Question ID : 444792657

Option 1 ID : 4447922236

Option 2 ID : 4447922238

Option 3 ID : 4447922237

Option 4 ID : 4447922235

Q.63 The correct order of C, N, O and F in terms of second ionisation potential is

- Options
1. $F < N < C < O$
 2. $C < O < N < F$
 3. $C < F < N < O$
 4. $C < N < F < O$

Question Type : MCQ

Question ID : 444792658

Option 1 ID : 4447922239

Option 2 ID : 4447922240

Option 3 ID : 4447922241

Option 4 ID : 4447922242

Q.64 "X" is an oxoanion of the lightest element of group 7 (in the periodic table). The metal is in +6 oxidation state in "X". The color of the potassium salt of X is

- Options
1. purple
 2. yellow
 3. orange
 4. green

Question Type : MCQ

Question ID : 444792660

Option 1 ID : 4447922248

Option 2 ID : 4447922249

Option 3 ID : 4447922250

Option 4 ID : 4447922247

Q.65 One mole of $\text{Cl}_2(\text{g})$ was passed into 2 L of cold 2M KOH solution. After the reaction, the concentrations of Cl^- , ClO^- and OH^- are respectively (assume volume remains constant)

- Options
1. 0.75M, 0.75M, 1M
 2. 0.5M, 0.5M, 1M
 3. 0.5M, 0.5M, 0.5M
 4. 1M, 1M, 1M

Question Type : MCQ

Question ID : 444792651

Option 1 ID : 4447922214

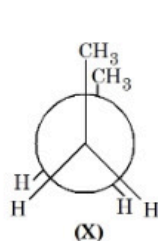
Option 2 ID : 4447922212

Option 3 ID : 4447922213

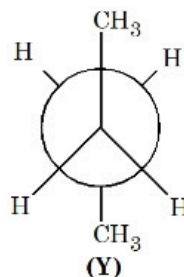
Option 4 ID : 4447922211

Q.66 Given below are two statements:

Statement I: There are several conformers for n-butane. Out of those conformers,



is the least stable and most stable conformer is



Statement II: As the dihedral angle increases, torsional strain decreases from (X) to (Y).

In the light of the above statements, choose the **correct** answer from the options given below

- Options
1. Statement I is false but Statement II is true
 2. Both Statement I and Statement II are true
 3. Statement I is true but Statement II is false
 4. Both Statement I and Statement II are false

Question Type : MCQ

Question ID : 4447922663

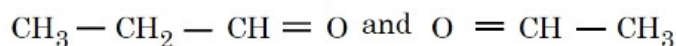
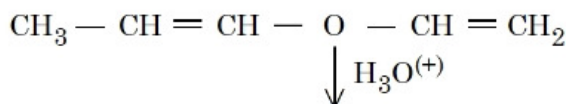
Option 1 ID : 4447922262

Option 2 ID : 4447922259

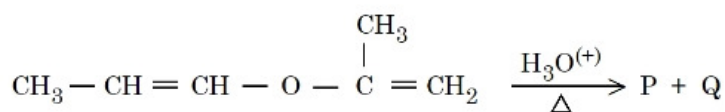
Option 3 ID : 4447922261

Option 4 ID : 4447922260

Q.67 The unsaturated ether on acidic hydrolysis produces carbonyl compounds as shown below:-



Based on this, predict the solution/reagent that will help to distinguish "P" and "Q" obtained in the following reaction:-



- Options
1. Lucas reagent
 2. Saturated NaHSO_3 solution
 3. Fehling solution
 4. 2, 4 - DNP reagent

Question Type : MCQ

Question ID : 4447922665

Option 1 ID : 4447922270

Option 2 ID : 4447922267

Option 3 ID : 4447922268

Option 4 ID : 4447922269

Q.68

Given below are two statements:

Statement I: The dipole moment of R-CN is greater than R-NC and R-NC can

undergo hydrolysis under acidic medium to produce $\text{R} - \overset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{OH}$.

Statement II: R-CN hydrolyses under acidic medium to produce a compound which on treatment with SOCl_2 , followed by the addition of NH_3 gives another compound(x). This compound (x) on treatment with NaOCl/NaOH gives a product, that on treatment with $\text{CHCl}_3/\text{KOH}/\Delta$ produces R-NC

In the light of the above statements, choose the *correct* answer from the options given below

- Options
1. Statement I is true but Statement II is false
 2. Both Statement I and Statement II are true
 3. Both Statement I and Statement II are false
 4. Statement I is false but Statement II is true

Question Type : MCQ

Question ID : 444792668

Option 1 ID : 4447922281

Option 2 ID : 4447922279

Option 3 ID : 4447922280

Option 4 ID : 4447922282

Q.69 The wavelength of spectral line obtained in the spectrum of Li^{2+} ion, when the transition takes place between two levels whose sum is 4 and difference is 2, is

- Options
1. $2.28 \times 10^{-7} \text{ cm}$
 2. $2.28 \times 10^{-6} \text{ cm}$
 3. $1.14 \times 10^{-7} \text{ cm}$
 4. $1.14 \times 10^{-6} \text{ cm}$

Question Type : MCQ

Question ID : 444792652

Option 1 ID : 4447922218

Option 2 ID : 4447922215

Option 3 ID : 4447922217

Option 4 ID : 4447922216

Q.70 Given below are two statements:

Statement I: Cross aldol condensation between two different aldehydes will always produce four different products.

Statement II: When semicarbazide reacts with a mixture of benzaldehyde and acetophenone under optimum pH, it forms a condensation product with acetophenone only.

In the light of the above statements, choose the *correct* answer from the options given below

- Options
1. Statement I is true but Statement II is false
 2. Both Statement I and Statement II are true
 3. Statement I is false but Statement II is true
 4. Both Statement I and Statement II are false

Question Type : MCQ

Question ID : 444792664

Option 1 ID : 4447922265

Option 2 ID : 4447922263

Option 3 ID : 4447922266

Option 4 ID : 4447922264

Section : Chemistry Section B

Q.71 Grignard reagent RMgBr (P) reacts with water and forms a gas (Q). One gram of Q occupies 1.4 dm^3 at STP. (P) on reaction with dry ice in dry ether followed by H_3O^+ forms a compound (Z). 0.1 mole of (Z) will weigh _____ g. (Nearest integer)

Question Type : SA

Question ID : 444792672

Q.72 Molar conductivity of a weak acid HQ of concentration 0.18 M was found to be $1/30$ of the molar conductivity of another weak acid HZ with concentration of 0.02 M. If $\lambda^\circ_{\text{Q}^-}$ happened to be equal with $\lambda^\circ_{\text{Z}^-}$, then the difference of the pK_a values of the two weak acids ($\text{pK}_a(\text{HQ}) - \text{pK}_a(\text{HZ})$) is _____ (Nearest integer).
[Given: degree of dissociation (α) $\ll 1$ for both weak acids, λ° : limiting molar conductivity of ions]

Question Type : SA

Question ID : 444792674

Q.73 The half-life of ^{65}Zn is 245 days. After x days, 75% of original activity remained. The value of x in days is _____. (Nearest integer)
(Given: $\log 3 = 0.4771$ and $\log 2 = 0.3010$)

Question Type : SA

Question ID : 444792675

Q.74 A chromium complex with a formula $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ has a spin only magnetic moment value of 3.87 BM and its solution conductivity corresponds to 1 : 2 electrolyte. 2.75 g of the complex solution was initially passed through a cation exchanger. The solution obtained after the process was reacted with excess of AgNO_3 . The amount of AgCl formed in the above process is ____ g. (Nearest integer)

[Given: Molar mass in g mol^{-1} Cr : 52; Cl: 35.5, Ag:108, O:16, H:1]

Question Type : SA

Question ID : 444792671

Q.75 0.25 g of an organic compound "A" containing carbon, hydrogen and oxygen was analysed using the combustion method. There was an increase in mass of CaCl_2 tube and potash tube at the end of the experiment. The amount was found to be 0.15 g and 0.1837 g, respectively. The percentage of oxygen in compound A is ____ %. (Nearest integer)

(Given: molar mass in g mol^{-1} H : 1, C : 12, O : 16)

Question Type : SA

Question ID : 444792673